1 Aligned equations

The trace $\operatorname{Tr} A$ of a matrix and $\arg\max_x f(x)$ in text, with $\mathbb{R}$, $\mathbb{N}$, $\mathfrak{g}$ and $\leqslant$, $\varnothing$.

$$a = b + c \tag{1}$$
$$d = e + f + g$$
$$h \leqslant \sum_{i=1}^{n} x_i^2 \tag{2}$$

Equation (**??**) is the first one.

$$x + y = z \tag{3}$$
$$\int_0^1 f(t)\,dt = F(1) - F(0) \tag{4}$$
$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{otherwise.} \end{cases} \tag{5}$$
$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{vmatrix} \alpha & \beta \\ \gamma & \delta \end{vmatrix}$$

$$p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + a_6 x^6 + a_7 x^7 + a_8 x^8 \tag{6}$$

$$\lim_{n\to\infty} \frac{1}{n} = 0, \qquad \frac{1}{2} + \tfrac{1}{3} = \frac{5}{6},$$
$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}, \qquad \sqrt[3]{8} = 2.$$